

1. (5 分) Answer the following questions:

(1) Please write down three types of power diodes.

(2) Which type are Thyristor and IGBT, current-driven device or voltage-driven device?

(3) Why the MOSFET can be easily paralleled to conduct higher current .

(1) 功率二极管分哪三种类型? (2 分)

(2) 分别说出晶闸管和 IGBT 属于电压控制还是电流控制型器件? (2 分)

(3) 为什么 MOSFET 器件容易实现并联均流? (1 分)

答案:

(1) 功率二极管分为普通整流二极管、快恢复二极管、肖特基二极管。

(2) 晶闸管属于电流控制型器件, IGBT 属于电压控制型器件。

(3) MOSFET 能够并联使用提高通流能力,是因为 MOSFET 的通态电阻具有正的温度系数,即通态电阻随温度升高为增大,因此具有均流的作用(电流大→温度升高→电阻增大→电流减小)。

2. (4 分) The switching waveforms of a power electronics device are shown in Fig.1. If $V_o=50V$, $I_o=10A$, on-state resistance is $20m\Omega$. The turn-on delay and turn off delay time are neglected.

Answer the following questions:

某一电力电子器件的开关波形如图 1 所示。如果 $V_o=50V$, $I_o=10A$, 通态电阻为 $20m\Omega$ 。忽略开通延迟和关断延迟时间。回答以下问题。

(1) What is the switching power loss of the switch? 开关管的开关损耗是多少? (2 分)

(2) What is the total power loss of the switch? 开关管的总损耗是多少? (2 分)

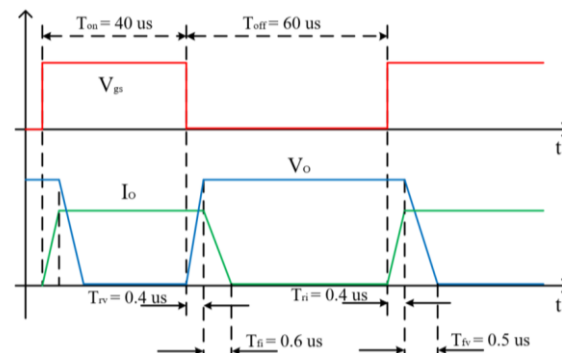


Fig. 1 Switching waveforms of a power electronics device

答案:

$$(1) P_s = \frac{V_o I_o f_s (T_{rv} + T_{fi})}{2} + \frac{V_o I_o f_s (T_{ri} + T_{fv})}{2}$$

$$f_s = \frac{1}{T_s} = 10 \text{ kHz}$$

$$P_s = 4.75 \text{ W}$$

$$(2) P_{on} = 0.782 \text{ W}$$

$$P = P_s + P_{on} = 5.532 \text{ W}$$

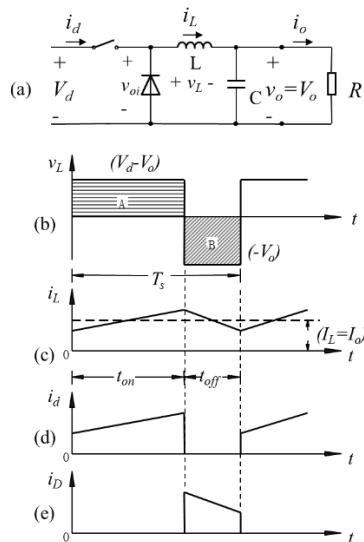
3. (8 分) In a Buck converter, consider all components to be ideal. Let input voltage V_d be 50~80V, output voltage $V_o = 40V$, and $L=60\mu H, f_s = 40kHz$. 在 Buck 变换器中, 考虑所有元件都是理想的。设输入电压 V_d 为 50~80V, 输出电压 V_o 为 40V, $L=60\mu H$, 开关频率为 40kHz。

- (1) Draw the Buck converter circuit diagram. 画出 Buck 变换器电路图。 (1 分)
- (2) Draw the waveforms of v_L , i_L , i_C . 画出 v_L , i_L , i_C 的波形。 (2 分)
- (3) Calculate the minimum output power P_{omin} that will keep the converter operating in a continuous-conduction mode for all range of V_d . 计算全输入电压范围内能够使变换器工作在 CCM 模式下的最小输出功率 P_{omin} 。 (3 分)
- (4) What will happen if keep $D=0.5$ and the load R is open? 如果占空比固定而负载突然开路, 会出现什么结果? (1 分)
- (5) Which is big for the Switch utilization in condition of $D=0.2$ and $D=0.8$? 比较在 $D=0.2$ 和 $D=0.8$ 两种情况下哪个开关利用率高? (1 分)

答案:

(1) Buck 变换器电路图:

(2) v_L , i_L , i_C 的波形:



(3) 计算最小输出功率 P_{omin} :

$$V_o = DV_d, V_d = 50 \sim 80V, V_o = 40V, D = 0.5 \sim 0.8,$$

$$V_o = 40V, L = 60\mu H, T_s = 25\mu s, \text{ 取最小值 } D = 0.5,$$

$$I_{oB} = I_{LB} = \frac{T_s V_o}{2L} (1 - D)$$

$$\text{代入得 } I_{oB} = 4.16A,$$

$$P_{omin} = V_o \cdot I_{oB} = 166.6W$$

(4) 如果占空比固定而负载突然开路, 输出电压会升高, 直至输出电压等于输入电压;

(5) $D=0.8$ 时开关利用率高。

4. (8 分) In a Buck-Boost converter, consider all components to be ideal. Let input voltage $V_d = 24V$, $D=0.6$, and $L=100\mu H, f_s = 50kHz, P_o=360W$. 在 Buck-Boost 变换器中, 假设所有元

件都是理想的。设输入电压 $V_d=24V$ ，占空比 $D=0.6$ ，电感为 $100\mu H$ ， $f_s=50kHz$ ， $P_o=360W$ 。

(1) Draw the Buck-Boost converter circuit diagram. 画出 Buck-Boost 变换器电路图。(1 分)

(2) Derive the relationship between the output voltage V_o and input voltage V_d , when the inductor current is continuous. 推导电感电流连续时输出电压 V_o 与输入电压 V_d 之间的关系式。(2 分)

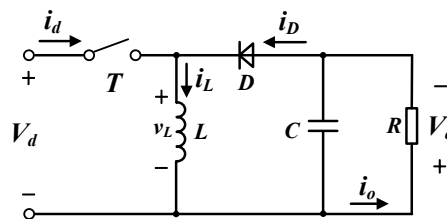
(3) If the output voltage ripple ratio $\Delta V_o/V_o$ is less than 2%, please calculate the minimum value of C in this converter. 如果输出电压纹波比 $\Delta V_o/V_o$ 小于 2%，请计算该变换器中电容 C 的最小值。(2 分)

(4) What is the voltage stress of the switch? 开关管的电压应力是多少？(1 分)

(5) What is the advantage and disadvantage of this converter? 该变换器的优缺点是什么？(2 分)

答案：

(1) Buck-Boost 变换器电路图：



(2) 输出电压 V_o 与输入电压 V_d 的关系式根据伏秒平衡推导：

$$V_d D T_s + (-V_o) (1 - D) T_s = 0$$

$$\therefore \frac{V_o}{V_d} = \frac{D}{1 - D}$$

(3)

$$\frac{V_o}{V_d} = \frac{D}{1 - D} \quad V_o=36V, \quad \Delta V_o/V_o < 2\%, \quad \Delta V_o=0.72V$$

$$P_o=360W, \quad V_o=36V, \quad I_o=10A, \quad T_s=20\mu s, \quad D=0.6,$$

$$\Delta V_o = \frac{\Delta Q}{C} = \frac{I_o D T_s}{C}$$

$$\text{代入得 } C_{\min} = 166.6 \mu F$$

(4) 开关管的电压应力： $V_T = V_d + V_o = 24 + 36 = 60V$

(5) 优点：电路可升压可降压。缺点：输入和输出反极性，开关管的电压、电流应力较大。